NETWORKING ASSIGNMENT

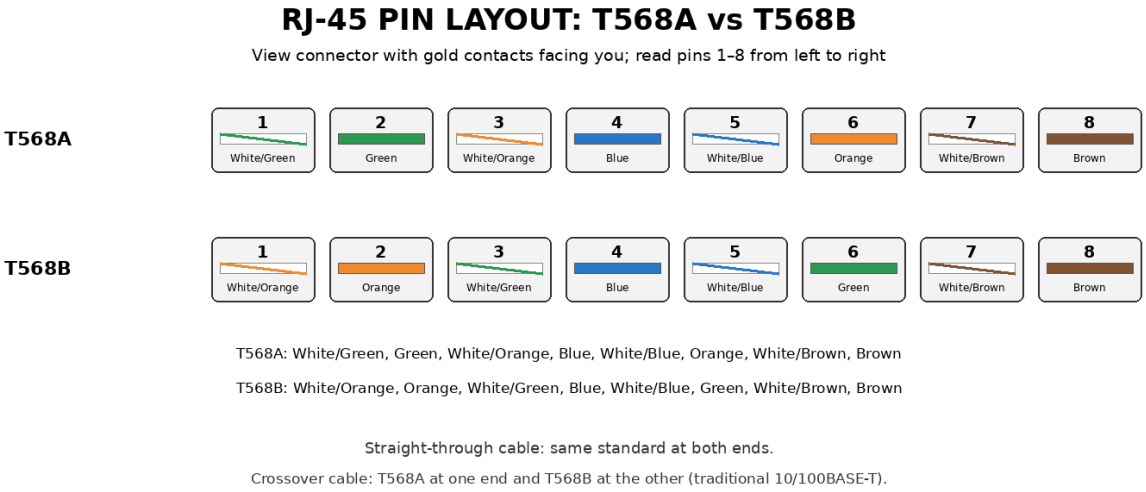
Ethernet Wiring Standards and Ethernet Technologies

TASK 1: RJ-45 Pin Layout and Wiring Standards

QUESTION

Draw the pin layout of an RJ-45 connector and label the wire colour order for both T568A and T568B.

ANSWER



Pin	T568A colour	T568B colour
1	White/Green	White/Orange
2	Green	Orange
3	White/Orange	White/Green
4	Blue	Blue
5	White/Blue	White/Blue
6	Orange	Green
7	White/Brown	White/Brown
8	Brown	Brown

For a straight-through cable, use the same standard at both ends. A traditional crossover cable uses T568A at one end and T568B at the other.

TASK 2: Compare Ethernet Standards

QUESTION

Compare 10BASE-T, 100BASE-TX, and 1000BASE-T by speed, cable type, maximum segment length, and connector.

ANSWER

Standard	Speed	Typical cable	Maximum length	Connector
10BASE-T	10 Mbps	Cat3 or better twisted-pair	100 m	RJ-45 (8P8C)
100BASE-TX	100 Mbps	Cat5 or better twisted-pair (Cat5e commonly used)	100 m	RJ-45 (8P8C)
1000BASE-T	1 Gbps (1000 Mbps)	Cat5e or better twisted-pair	100 m	RJ-45 (8P8C)

The 100 m limit is the usual maximum Ethernet channel/segment length for these copper twisted-pair standards under compliant cabling conditions.

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TASK 3: Direct Ethernet Connection Between Two Laptops

QUESTION

You need to connect two laptops directly using Ethernet, without a network switch. Choose a straight-through or crossover cable and explain how the transmit and receive pairs affect your choice.

ANSWER

Traditionally, a crossover cable is used to connect two similar devices directly, such as two laptops with 10/100 Ethernet ports that do not support Auto-MDI/MDI-X. It swaps the transmit and receive pairs so that one laptop’s transmit pair connects to the other laptop’s receive pair.

However, many modern Ethernet ports support Auto-MDI/MDI-X, which automatically adjusts transmit and receive pairs. With these devices, a normal straight-through cable usually works as well. If the laptop ports’ capabilities are unknown, a crossover cable is the traditional answer for this assignment.

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TASK 4: Ethernet Devices at Home or College

QUESTION

Name three devices commonly found at home or college that may use Ethernet. For each, state a likely supported Ethernet standard and note that the exact capability depends on the device model.

ANSWER

Device	Likely Ethernet standard	Reason
Wi-Fi router LAN port	1000BASE-T on many current models; some support 100BASE-TX	Gigabit Ethernet is common on newer routers, but entry-level models may have
Desktop computer or docking station	1000BASE-T	Gigabit Ethernet is widely supported on desktop network adapters and many
Network printer or college network switch	100BASE-TX or 1000BASE-T	Many printers use 100 Mbps Ethernet, while modern switches commonly pro

Check the device label, manual, or network-adapter settings to confirm its actual supported speed.

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